

Python Assignment – 3

Task 1: While Loop & Control Statements -Number Guessing Game.

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[3] #Task 1: While Loop & Control Statements -Number Guessing Game
import random

# Generate random number between 1 and 10
secret_number = random.randint(1, 10)

attempts = 3

while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))

    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue

    attempts -= 1

    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
    elif guess > secret_number:
        print("Too high. Try again.")
    else:
        print("Too low. Try again.")
else:
    print("Better luck next time!")
```

Task 2:- For Loop - Multiplication Table Generator.

```
#Task 2:- For Loop - Multiplication Table Generator
number = int(input("Enter the number for which you want the multiplication table: "))

for i in range(1, 11):
    result = number * i
    print(f"{number} x {i} = {result}")
```

```
Enter the number for which you want the multiplication table: 9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
```

Task 3: Function- BMI Calculator.

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#Task 3: Function- BMI Calculator
def calculate_bmi(weight, height):
    bmi = weight / (height ** 2)
    return bmi

weight = float(input("Enter your weight in kg: "))
height = float(input("Enter your height in meters: "))

bmi = calculate_bmi(weight, height)
print(f"Your BMI is: {bmi:.2f}")

... Enter your weight in kg: 80
Enter your height in meters: 1.45
Your BMI is: 38.05
```